

Document Type Definition (DTD) in XML – Detailed Notes

📌 What is DTD?

- DTD (**Document Type Definition**) defines the **structure** and the **rules** for an XML document.
 - It lists:
 - **Elements** that can appear in the document
 - **Attributes** for each element
 - **Order and nesting** of elements
 - Whether elements can be empty or contain text or other elements
 - Ensures that the XML document is **valid** (follows the defined structure).
-

📌 Why use DTD?

- To **validate** XML data structure.
 - To **ensure consistency** across multiple XML documents.
 - To act as a **blueprint** for developers and parsers on how the XML document should be structured.
 - Useful in **data sharing**, especially in web services and system integrations.
-

📌 Types of DTD

1. Internal DTD

- Defined **within the XML document itself**.
- Useful for **small or self-contained** documents.

📌 Example:

```
<?xml version="1.0"?>
<!DOCTYPE note [
  <!ELEMENT note (to, from, heading, body)>
  <!ELEMENT to (#PCDATA)>
  <!ELEMENT from (#PCDATA)>
  <!ELEMENT heading (#PCDATA)>
  <!ELEMENT body (#PCDATA)>
]>
<note>
  <to>John</to>
  <from>Mary</from>
  <heading>Reminder</heading>
  <body>Don't forget the meeting!</body>
</note>
```

2. External DTD

- Defined in a **separate file** with `.dtd` extension.
- Can be **reused** by multiple XML files.
- Referenced using `SYSTEM` keyword.

📌 XML File:

```
<?xml version="1.0"?>
<!DOCTYPE note SYSTEM "note.dtd">
<note>
  <to>John</to>
  <from>Mary</from>
  <heading>Reminder</heading>
  <body>Meeting at 10 AM</body>
</note>
```

❏ note.dtd File:

```
<!ELEMENT note (to, from, heading, body)>
<!ELEMENT to (#PCDATA)>
<!ELEMENT from (#PCDATA)>
<!ELEMENT heading (#PCDATA)>
<!ELEMENT body (#PCDATA)>
```

XML DTD – Formal Public Identifier (FPI) and Components

❏ What is FPI (Formal Public Identifier)?

FPI stands for **Formal Public Identifier**, a unique identifier used to refer to a **public DTD**. It allows XML parsers to **find and use external DTDs** stored publicly, not necessarily on the local machine.

❏ Purpose of FPI:

- To **identify** and **reuse** standard DTDs.
- To **distinguish** between different versions of the same DTD.
- To provide a **consistent way** of referencing shared DTDs.

❏ Syntax of FPI:

```
`<!DOCTYPE root-element PUBLIC "FPI" "URI">`
```

Where:

- "FPI" = Formal Public Identifier (standard naming)
- "URI" = System Identifier (URL or path where the DTD is located)

❏ Example:

```
`<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
```

Parts of the FPI:

Part	Description
-	Indicates the owner is not ISO (e.g., W3C, not an official standards body)
//W3C	Organization that defined the DTD
//DTD XHTML 1.0 Strict	Type and name of the document
//EN	Language (EN = English)

⌘ FPI vs SYSTEM:

Type	Keyword	Description
Public DTD	PUBLIC	Uses an FPI and optionally a URL
Private DTD	SYSTEM	Refers directly to a URL or local file path

Example – SYSTEM:

```
` <!DOCTYPE note SYSTEM "note.dtd">`
```